Test 2: Temporal Representation & Feedforward Transformations — Solutions

Total: 20 marks plus 2 bonus. Simulation data: `tests_data/make_data.py`.
Mark splits are inferred from the marked paper where possible.

Question 1 (4)

(a) [2]

1. **Frequency grid.** With time step Δt, use $N = T/\Delta t$ samples. The discrete frequencies are $\omega_k = 2\pi k/T$.
2. **Sample the band.** For every $0 < |\omega_k| \le 2\pi L$, draw $X(\omega_k) = a + ib$ with $a, b \sim \mathcal{N}(0, 1)$. Set every other coefficient to 0, including $\omega = 0$ (zero mean).
3. **Make it real.** Set $X(-\omega_k) = X^*(\omega_k)$ (see (b)).
4. **Back to time and rescale.** Take the inverse FFT, $x(t) = \mathcal{F}^{-1}[X]$. Then rescale, $x \leftarrow x \cdot \sigma / \mathrm{RMS}(x)$, so that the RMS is exactly σ.

Marking

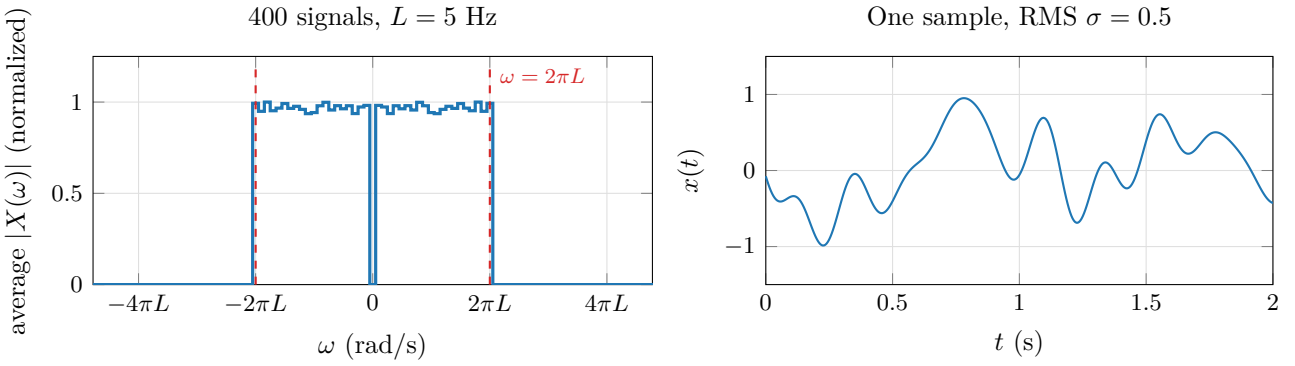
0.5 each: grid or band in rad/s ($2\pi L$), random complex coefficients inside the band and zero outside, symmetry, inverse FFT *and* rescale to RMS σ. A low-pass "smoothing filter" in place of the inverse FFT and rescale loses the last half-mark.

(b) [1]

$$X(-\omega) = X^*(\omega),$$

so the real part is even and the imaginary part is odd, and $X(0)$ must be real. Each pair $\pm\omega$ then sums to a real cosine with a phase, $2|X|\cos(\omega t + \phi)$. The lecture notes write this loosely as "$X(\omega) = X(-\omega)$". The conjugate is the precise statement.

(c) [1] On average $|X(\omega)|$ is **flat** (white) inside the band and zero outside. On an angular-frequency axis the edges are at $\omega = \pm 2\pi L$. On an axis in Hz they are at $\pm L$.



The dip at $\omega = 0$ comes from setting $X(0) = 0$ (zero mean).

Marking

0.5: flat inside the band, zero outside, symmetric about 0. 0.5: axes labelled and the cutoff marked at $\pm 2\pi L$ on a ω axis (or $\pm L$ on a Hz axis). Writing $\pm L$ on an axis labelled ω loses this half-mark.

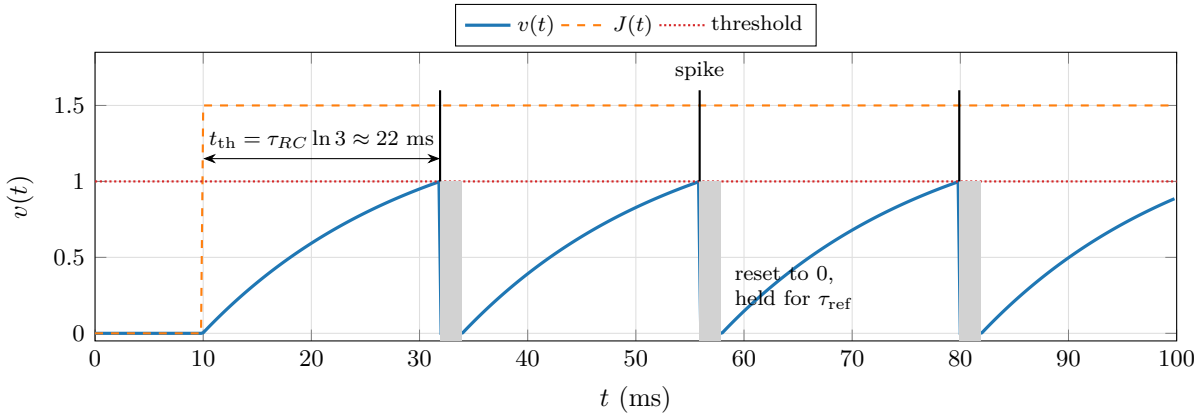
Question 2 (4)

- v: the (normalized) **membrane voltage**. The threshold is 1 and the reset value is 0.
- J: the (normalized) **input current** into the soma, in the same units, so $J = 1$ is the threshold current.
- $\tau_{RC} = RC$: the **membrane time constant**, from the membrane's capacitance C and leak resistance R. It sets how fast v moves.
- dv/dt: the rate of change of the voltage. The $-v$ term is the **leak**: charge drains out at a rate proportional to v.

The whole equation is a leaky integrator (an RC circuit). The voltage charges up and relaxes exponentially toward J with time constant τ_{RC}: from $v(0) = 0$, $v(t) = J(1 - e^{-t/\tau_{RC}})$.

- If $J \leq 1$, v settles at J below threshold and the neuron never fires.
- If $J > 1$, v reaches the threshold 1 after $t_{\text{th}} = -\tau_{RC} \ln(1 - 1/J)$. At threshold the neuron **emits a spike**, v is **reset to** 0 and **held there for the refractory period** τ_{ref}, and then integration restarts.

This repeats with period $\tau_{\text{ref}} + t_{\text{th}}$, which gives the rate $G[J] = \left(\tau_{\text{ref}} - \tau_{RC} \ln(1 - 1/J)\right)^{-1}$. The plot below uses $\tau_{RC} = 20$ ms and $\tau_{\text{ref}} = 2$ ms, with J stepping from 0 to 1.5 at $t = 10$ ms.



Marking

0.5 each for v, J, τ_{RC}, and dv/dt (2). 1: leaky integration toward J (RC circuit), including no spikes when $J \leq 1$. 1: at threshold, spike, *reset to* 0, and *refractory period* τ_{ref}. Missing the reset or refractory period loses half a mark.

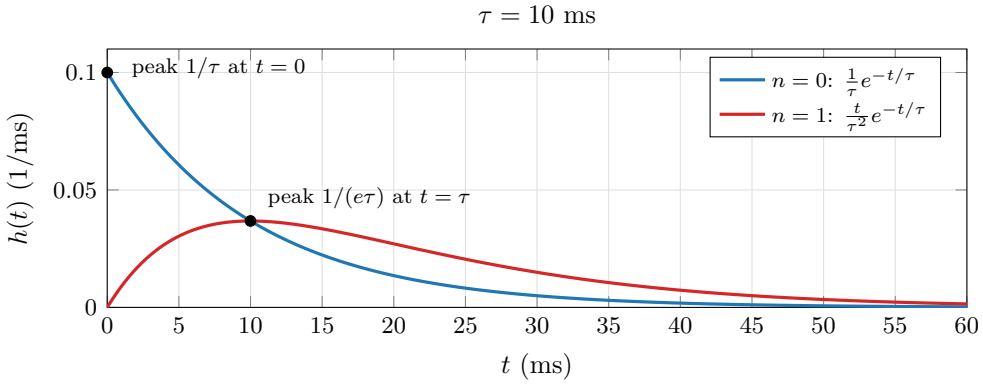
Question 3 (4)

(a) [2] The normalization is $c = \int_0^\infty t^n e^{-t/\tau} dt = n!\,\tau^{n+1}$.

- $n = 0$: $h(t) = \frac{1}{\tau} e^{-t/\tau}$. This is the exponential synapse. It jumps to its maximum $1/\tau$ **at** $t = 0$ and then decays.

- $n = 1$: $h(t) = \frac{t}{\tau^2} e^{-t/\tau}$. This is the alpha synapse. It **starts at** 0, rises to a peak of $1/(e\tau) \approx 0.37/\tau$ at $t = \tau$, and has a longer tail.

Both have unit area.

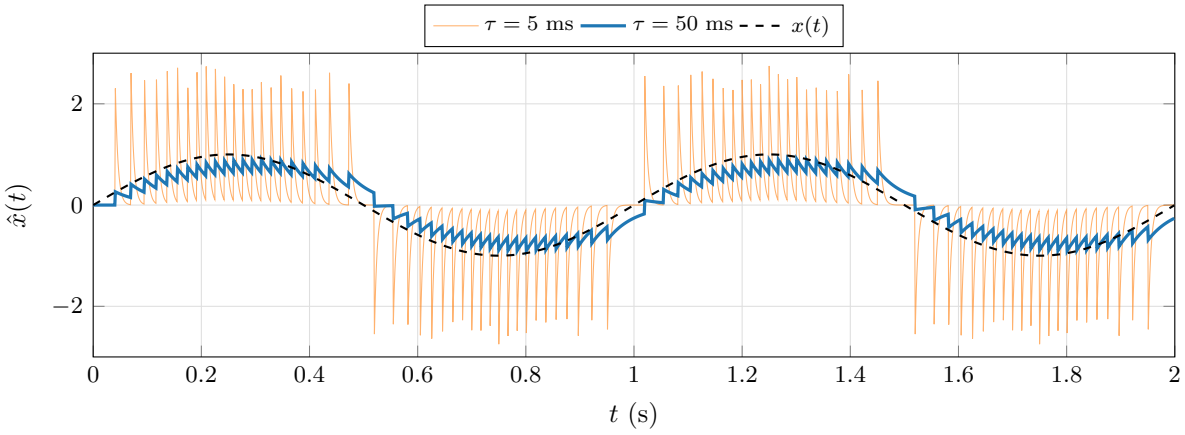


Marking

1: $n = 0$ curve has its maximum at $t = 0$ and decays monotonically. 1: $n = 1$ curve starts at 0, peaks later (at $t = \tau$), and is lower and wider.

(b) [2] A larger τ is a lower cut-off low-pass filter (cut-off $\approx 1/(2\pi\tau)$ Hz). Two effects on $\hat{x}(t)$:

1. **Smoother.** Spike noise (the ripple from individual spikes) is averaged over a longer window, so $\hat{x}$ is less noisy.
2. **More delayed and sluggish.** $\hat{x}$ lags x and responds more slowly to fast changes. Fast signal components are attenuated.



This uses the two LIF neurons from Question 5, decoded with the $n = 0$ filter. With $\tau = 50$ ms the trace is much smoother, but it is visibly shifted right (for a 1 Hz sine the lag is about $\arctan(2\pi \cdot 0.05)/2\pi \approx 48$ ms).

Marking

1 each: smoother or less noisy; more delay or lag (slower response to changes). "Lower amplitude for fast signals" also earns the second mark.

Question 4 (4)

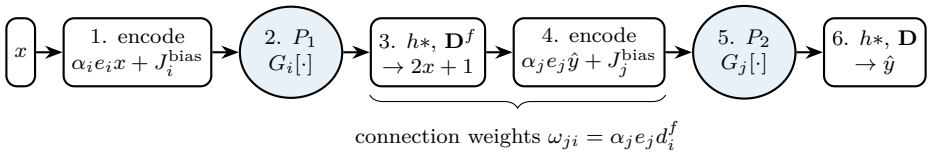
(a) [1]

- **Function decoders on** P_1 for $f(x) = 2x + 1$: $\mathbf{D}^f = \left((\mathbf{A}\mathbf{A}^\mathsf{T} + N\sigma^2\mathbf{I})^{-1}\mathbf{A}\mathbf{Y}^\mathsf{T}\right)^\mathsf{T}$, with $\mathbf{Y} = f(\mathbf{X})$ evaluated at the sample points.
- **Identity decoders on** P_2 (target y), to read out the final $\hat{y}$.

The $+1$ is affine, so identity decoders on P_1 with a transform of 2 cannot produce it alone. Either decode $2x+1$ directly or add a constant input. Also, if $x \in [-1, 1]$ then $y \in [-1, 3]$, so P_2 needs a radius of about 3.

(b) [3]

1. **Encode into** P_1**.** The input x becomes a current in each neuron: $J_i = \alpha_i \langle e_i, x\rangle + J_i^{\text{bias}}$.
2. P_1 **nonlinearity.** Each neuron spikes: $a_i(t) = G_i[J_i(t)]$.
3. **Decode** f **from** P_1**.** The spikes are filtered by the synapse and weighted by the function decoders: $\hat{y} = \sum_i d_i^f \, (h * a_i) \approx 2x + 1$.
4. **Encode into** P_2**.** $J_j = \alpha_j \langle e_j, \hat{y}\rangle + J_j^{\text{bias}}$. Steps 3 and 4 together are the weight matrix $\omega_{ji} = \alpha_j \langle e_j, d_i^f\rangle$.
5. P_2 **nonlinearity.** $b_j(t) = G_j[J_j(t)]$.
6. **Decode the output.** Filter P_2's spikes and apply its identity decoders: $\hat{y} = \sum_j d_j \, (h * b_j)$.



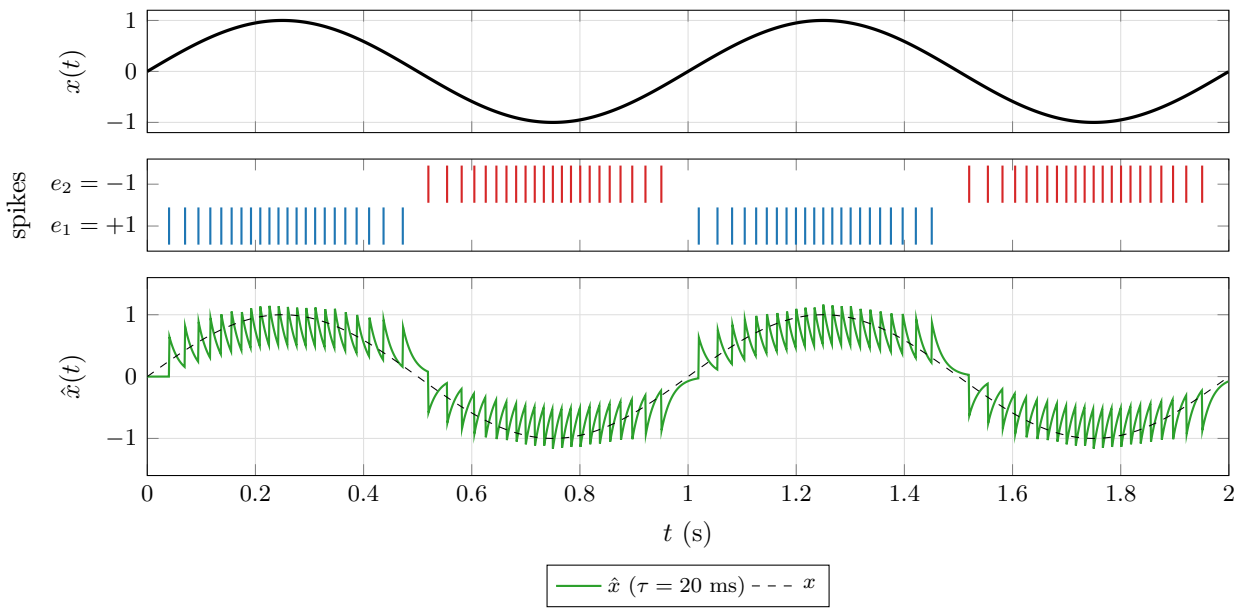
Marking

(a) 0.5: function decoders for $2x + 1$ on P_1. 0.5: identity decoders on P_2 for the output. (b) 0.5 per step, in the correct order (encode, nonlinearity, decode with filter, twice).

Question 5 (4)

Same α and J^{bias} with opposite encoders gives mirror-image tuning curves. Neuron 1 fires only when x is above its intercept, and neuron 2 only when x is below the mirrored intercept. In the simulation below the intercepts are at 0 and $a^{\max} = 60$ Hz.

- **Spike trains.** Neuron 1 fires during the positive half-cycles, with the **highest spike density at the peaks** ($x = +1$) and sparse spikes near the zero crossings. Neuron 2 does the same during the negative half-cycles, densest at the troughs. Neither fires near $x = 0$.
- **Decoded signal.** With $d_2 = -d_1$, each neuron-1 spike adds a filtered bump of $+d_1$ and each neuron-2 spike adds one of $-d_1$. $\hat{x}(t)$ follows the sine in shape and size, but it is **noisy**: a sawtooth ripple from individual spikes. It is slightly **delayed** by the synapse.



Marking

1: sine input over a couple of periods. 1: each neuron fires only in its own half-cycle. 0.5: spike density increases toward the peak or trough (rate code). 1: $\hat{x}$ tracks the sine and is drawn as a noisy, rippled reconstruction rather than a clean sine. 0.5: slight delay, or near-zero output where neither neuron fires.

Where marks are usually lost

Drawing evenly spaced spikes (the rate must follow $|x|$), or drawing $\hat{x}$ as a perfect sine. Two neurons give a very rough estimate.

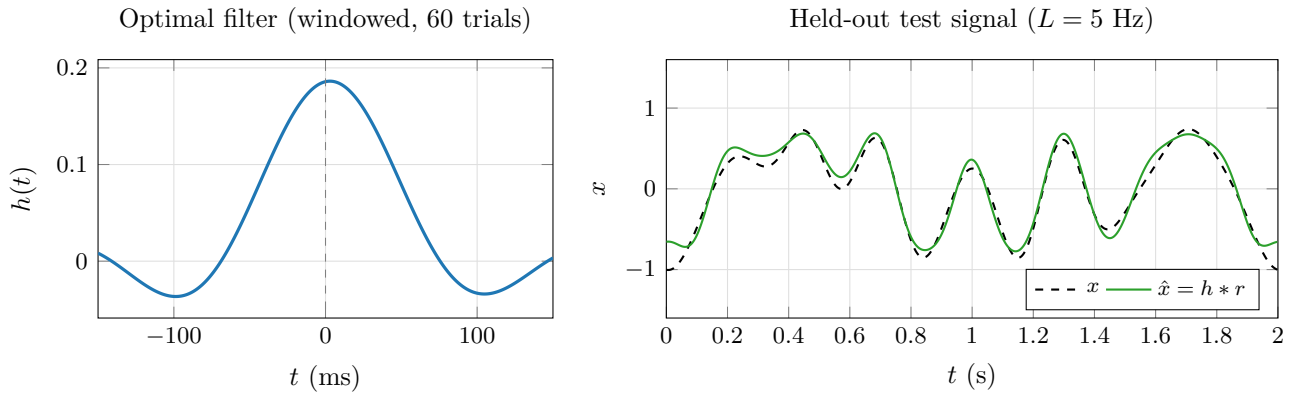
Bonus Question 1 (2)

$H(\omega)$ is the least-squares (Wiener) filter. It is the linear filter h for which $\hat{x} = h * r$ is as close as possible to x. At each frequency it acts like a regression coefficient:

$$H(\omega) = \frac{\text{cross-spectrum (covariance) of signal and response}}{\text{power spectrum (variance) of the response}}.$$

- **Numerator** $\langle X R^* \rangle$: how strongly, and with what phase, the response at frequency ω is correlated with the signal. Spike power that is unrelated to the signal averages out to about zero over many trials or windows.
- **Denominator** $\langle |R|^2 \rangle$: the total response power at ω, both signal-related and spike noise. Dividing by it normalizes the gain. Frequencies where the response is mostly spike noise get a small gain, and frequencies where the response reliably tracks the signal get a large one.

The phase of H corrects for the delay between signal and spikes, so $h(t)$ is generally **non-causal** (nonzero for $t < 0$). The lecture's windowed version smooths both spectra in frequency, which gives a smoother, shorter $h(t)$.



Here $r = s_1 - s_2$ is the signed spike train of the two neurons from Question 5.

Marking

1: the filter finds the linear filter that minimizes the squared decoding error, by correlating input with response across frequencies. 1: numerator is the signal–response cross-correlation (cross-spectrum), and the denominator normalizes by the response power, so noise-dominated frequencies are suppressed.